

4.1.2 Metabolism and exercise

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the consequences of exercise on the body	To include both immediate and long term effects on the respiratory and cardiovascular systems and on skeletal muscle. HSW12
(b) the factors affecting aerobic fitness	To include age, gender and participation in exercise.
(c) practical investigations into the effect of factors on (resting) heart rate, breathing rate or recovery times and the analysis of primary and secondary data	M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M1.9, M3.2, M3.5 PAG10 HSW3, HSW4, HSW5, HSW6, HSW12

(d) (i) how training can significantly improve aerobic fitness	To include frequency, intensity, type and time (duration) (F.I.T.T. factors) of exercise programme. <i>M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M1.9, M3.5</i> PAG10 HSW4, HSW5, HSW6, HSW12
(ii) practical investigations into the effects of F.I.T.T. factors on (resting) heart rate, breathing rate or recovery times	
(e) supplementary methods of enhancing athletic performance	To include the description and evaluation of carbohydrate loading diets, the use of recombinant erythropoietin (RhEPO), blood doping and use of steroids.
(f) the significance of VO_2 max as a measure of aerobic fitness	To include considerations of VO_2 max as the maximum rate at which oxygen can be taken in, transported and utilised. <i>M0.1, M0.2, M1.9, M3.1</i> HSW1, HSW3, HSW4, HSW5, HSW6, HSW9, HSW10, HSW12
(g) the role of haemoglobin in oxygen transport	
(h) the oxygen dissociation curves for different respiratory pigments	To include the differences in affinity for oxygen between adult haemoglobin, fetal haemoglobin and myoglobin and the reasons for the differences. <i>M0.1, M0.2, M1.1, M1.2, M1.6, M3.1, M3.4, M3.6</i> HSW8
(i) the factors which affect oxygen dissociation from respiratory pigments	To include temperature, pH, carbon dioxide and the role of carbonic anhydrase.
(j) the build-up of an oxygen deficit and oxygen debt/EPOC (Excess Post-exercise Oxygen Consumption)	To include the differences between the oxygen deficit and oxygen debt.
(k) (i) the histology and ultrastructure of skeletal muscle	To include the interpretation of diagrams, photomicrographs and electron micrographs. <i>M0.1, M0.2, M1.1, M1.2, M1.8</i>
(ii) observations of muscle tissue made using a light microscope and muscle tissue responses to ATP and other solutions	To include muscle dissections e.g. chicken leg with thigh. <i>M0.1, M0.2, M1.1, M1.2, M1.8</i> PAG1 HSW4, HSW5, HSW6
(l) the sliding filament theory of muscle contraction.	To include the role of actin, myosin, troponin, tropomyosin, the importance of the power stroke and the role of ATP and calcium ions. HSW8